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Prepared By	Owner
Approved By	Jacob Honick, Owner
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QP-022 First Article Inspection

Standards and clause references are guidance only and do not claim certification or standard compliance. Applicable customer, contractual and legal requirements govern accepted work.

1. Purpose

Define the process for performing First Article Inspection (FAI) to verify that production processes are capable of producing parts that meet all drawing and specification requirements, in alignment with AS9102.

2. Scope

Applies when the customer order requires formal FAI. Full first-piece inspection and full final inspection remain the default under QP-006, WI-005, QP-007 and WI-006 even when a formal FAI package is not required.

3. Responsibilities

Role	Responsibility
Shop Owner	Determine when FAI is required, perform ballooning and measurement, complete FAI forms, submit to customer for approval when required, maintain FAI records

4. References

- QF-007 — FAI Form 1 (Part Number Accountability)
- QF-008 — FAI Form 2 (Product Accountability - Raw Material)
- QF-009 — FAI Form 3 (Characteristic Accountability)
- QP-015 — Configuration Management
- QP-025 — Control of Changes
- AS9102 — use the revision and reporting format required by the customer. A standard reference alone does not assert compliance or establish a formal FAI obligation.
- AS9100 Rev D Clause 8.5.1.1

5. Definitions

- **First Article Inspection (FAI):** A complete, independent, and documented physical and functional inspection process to verify that prescribed production methods have produced an acceptable item as specified by engineering drawings, planning, purchase order, specifications, and/or other applicable design documents.
- **Partial FAI:** A reduced FAI scope applied when changes affect only specific characteristics and previously verified characteristics remain unchanged.
- **Ballooning:** The process of numbering each characteristic on the engineering drawing to correspond with measurements recorded on the FAI forms.

6. Procedure

FAI Triggers

1. Review the order and applicable customer specifications for formal FAI requirements and triggers. Where imposed, apply the required standard revision, full/partial scope, production-break rules and customer approval hold points. New parts, design/process/source changes and interruptions prompt this review; they do not by themselves impose an AS9102 package when the customer has not required one.

Partial FAI Criteria

1. When formal FAI is required, a partial FAI may be performed only when permitted by the applicable customer requirements and required standard revision, and when:
    - A design change affects only specific characteristics AND
    - Previously verified characteristics and associated production methods remain unchanged AND
    - The customer has not required a full FAI

2. Document justification for partial FAI on Form 1. Reference the prior full FAI number.

Drawing Ballooning

1. Create a ballooned drawing: assign a sequential number to each unique characteristic (dimension, tolerance, note, surface finish, material callout) on the drawing.
2. Include ALL characteristics: dimensional, geometric tolerances, surface finishes, material specifications, process notes, and special requirements.

Characteristic Recording

1. Measure the first article part for every ballooned characteristic.
2. Record actual measured values on QF-009 (Form 3) for each characteristic number. Record the nominal value, tolerance, and actual measured value.
3. For material requirements: record material certification data on QF-008 (Form 2) including specification, grade, heat/lot number, and supplier.
4. Complete QF-007 (Form 1) with part number, drawing revision, FAI reason, partial FAI justification (if applicable), and sign-off.

Acceptance and Customer Approval

1. Review all measurements against drawing requirements. All characteristics must be within tolerance for FAI acceptance.
2. If any characteristic is out of tolerance, the FAI FAILS. Correct the process, produce a new first article, and re-inspect.
3. Submit FAI package (Forms 1, 2, 3 plus ballooned drawing) to the customer for approval when contractually required.
4. Do not proceed to production quantities until FAI is accepted (internal acceptance or customer approval as applicable).

7. Records Generated

Form No.	Title	When Completed
QF-007	FAI Form 1 - Part Number Accountability	At each FAI event
QF-008	FAI Form 2 - Product Accountability	At each FAI event
QF-009	FAI Form 3 - Characteristic Accountability	At each FAI event

8. Revision History

Rev	Date	Description	Approved By
A	2026-08-07	Initial release	Shop Owner

Revision B Approval

Approved 2026-11-05 by Jacob Honick, Owner. Apply formal FAI when required by the customer; retain full first-piece and acceptance inspection by default.